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Oracle Agent

The Oracle Agent runs your solution/solve.sh in the task environment and verifies it passes all tests. It's the first line of validation for your task.

The Oracle Agent will be run 3 times.

Getting Started

Video Tutorial

New video coming soon...

Practice Notebook

Download the Jupyter notebook for hands-on practice:

New Jupyter notebook coming soon...

What You'll Learn

- What the Oracle Agent is and how it works
- How to run the Oracle Agent on your task
- How to interpret Oracle output
- How to debug failing runs
- How to iterate and fix issues

What is the Oracle Agent?

The Oracle Agent is an automated agent that:

1. Starts your Docker environment
2. Executes your solution/solve.sh commands
3. Runs your tests to verify completion
4. Reports pass/fail results

If the Oracle Agent can't complete your task, neither can AI agents.

Key Concept: The Oracle Agent verifies that your task is **solvable**. If the Oracle can't solve your task, it's broken.

Running the Oracle Agent

BASH

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```
# Basic run
harbor run -a oracle -p <task-folder>

# With verbose output
harbor run -a oracle -p <task-folder> -v
```

Expected Output

Successful Run

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```
Starting task: my-task
Building Docker environment...
Running oracle solution...
  ✓ Step 1 completed
  ✓ Step 2 completed
  ✓ Step 3 completed
Running tests...
  ✓ test_output_exists PASSED
```

```
✓ test_format_correct PASSED
✓ test_values_valid PASSED
```

RESULT: PASS

Failed Run

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```
Starting task: my-task
Building Docker environment...
Running oracle solution...
  ✓ Step 1 completed
  x Step 2 failed: command not found

RESULT: FAIL
Error: Solution did not complete successfully
```

Debugging Failures

Debugging Workflow

When the Oracle Agent fails, follow this workflow:

Step 1: Identify the Failure

- Read the error message carefully
- Which step failed?
- What was the error?
- What file/command was involved?

Step 2: Reproduce Interactively

BASH

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```
harbor tasks start-env -p <task-folder> -i
```

Inside the container, run commands one by one to find the issue.

Step 3: Fix and Re-test

1. Update `solution/solve.sh` or `environment/Dockerfile`
2. Run Oracle again
3. Repeat until passing

Solution Fails

If your solution doesn't run:

1. **Enter interactive mode:**

BASH

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```
harbor tasks start-env -p <task-folder> -i
```

2. **Run commands manually** to find the failing step
3. **Check for:**
 - Typos in commands
 - Missing dependencies
 - Wrong file paths
 - Permission issues

Tests Fail

If the solution runs but tests fail:

1. **Check test output** for specific failures
2. **Verify your solution** actually produces expected output
3. **Check for:**
 - Incorrect output format
 - Missing files
 - Off-by-one errors

- Edge cases not handled

Environment Issues

If the container won't build:

1. **Check Dockerfile syntax** in `environment/Dockerfile`
2. **Verify base image** exists and is accessible
3. **Check dependency installation** commands
4. **Try building manually:**

BASH

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```
cd <task-folder>/environment
docker build -t test .
```

Common Issues

ISSUE	CAUSE	SOLUTION
"command not found"	Missing dependency	Add to environment/Dockerfile
"file not found"	Wrong path	Use absolute paths
"permission denied"	File permissions	Check chmod in environment/I
Tests timeout	Solution too slow	Optimize or increase timeout

Common Debugging Scenarios

Missing File:

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Error: No such file: /app/data/input.csv

Fix: Check `environment/Dockerfile` COPY commands and file paths

Command Not Found:

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Error: grep: command not found

Fix: Add package to `environment/Dockerfile` `apt-get install`

Test Assertion Failed:

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AssertionError: Expected 42, got 41

Fix: Debug your solution logic

Timeout:

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Error: Task exceeded timeout (1800s)

Fix: Optimize solution or increase timeout in `task.toml`
(`[agent].timeout_sec`)

Oracle vs Real Agents

ORACLE AGENT	REAL AGENTS (GPT-5.5, ETC.)
Runs your solution/solve.sh	Generate their own solution

ORACLE AGENT	REAL AGENTS (GPT-5.5, ETC.)
Always deterministic	May vary between runs
Tests task validity	Tests task difficulty
Must always pass	May fail (that's the goal!)

Best Practices

1. **Run oracle early and often** — Don't wait until submission
2. **Fix oracle failures first** — If oracle fails, task is broken
3. **Check all test output** — Understand why tests pass/fail
4. **Keep environment minimal** — Easier to debug

Practice Exercise

Using the practice notebook:

1. Load the sample task
2. Run the Oracle (it will fail)
3. Identify the issue
4. Fix the solution
5. Run Oracle again until passing

Next Steps

- [Run against real agents](#)
- [Review CI checks](#)

PREVIOUS

[Writing Tests](#)

NEXT

[NOP Agent](#)